INFO 6150 Web Design and User Experience Engineering

SPRING 25

Course Information

Course Title: Web Design and User Experience Engineering
Course Number: INFO 6150
Term and Year: Fall 2024
Credit Hour: 4
CRN: 31876
Course Format: On-Ground/Traditional

Instructor Information

Full Name: Vishal Chawla
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Phone: 339-293-2010
Office Hours: On Demand by Zoom or Before/After class

Instructor Biography

Vishal Chawla is the Director of Enterprise Collaboration, AI and Web Services at MIT Lincoln Labs, with over 25 years of professional experience. He holds a Master's degree in Information Technology from Northeastern University, where he specialized in Web Design, Web Development, User Experience, Application Development, Quality Assurance, Web Services, Database Management, Open Source Development, and GenAI.

Chawla is an esteemed adjunct faculty member at Northeastern University, teaching courses in Web Design and User Experience Engineering, as well as User Experience Design and Testing. He is recognized as an industry expert with a deep knowledge of technology and a strong track record in leadership roles. His work has significantly contributed to the development of innovative technological solutions and products.

In addition to his professional achievements, Chawla is passionate about teaching and mentoring. His interactive teaching style and dedication to student success have earned him high regard among his students. He is committed to fostering his students' potential and guiding them toward successful careers. A lifelong learner, Chawla believes in the importance of continuous self-improvement and adaptation in the ever-evolving field of technology.

Course Prerequisites

No Prerequisites Required

Course Description

Exposes students to both conceptual and technical aspects of Web design. User experience design is the discipline of creating a useful and usable website or application that is easily navigated and meets the needs

of both the site owner and its users. Covers Web standards and best practices. Studies the fundamental concepts, techniques, practices, workflows, and tools associated with the practice of user-experience design in Web interfaces. Additional areas of focus include typography, color theory and composition, responsive design, CSS3 concepts, basic scripting, and JavaScript libraries to create functional, effective, and visually appealing websites.

Course Learning Outcomes

Introduces front-end web design and principles concepts while working on web design projects. The course allows students to quickly understand and develop web pages that cater to customer and business needs. While doing the labs and final project, students can use and learn JavaScript skills, including libraries like Bootstrap, JQuery, React, Node JS, Express JS, and AngularJS. Security techniques will be discussed that can help avoid code injections in websites. The course will also introduce user experience concepts and their relation to front-end web design.

1. Understand the User Experience concepts and the importance of making usable websites or products
2. Introduce front-end web design concepts
3. Develop web pages quickly using HTML and JavaScript as per the business needs
4. Learn HTML and JavaScript concepts
5. Learn concepts of Bootstrap, JQuery, Angular JS, React JS, NodeJS, Express JS JavaScript library
6. Work in the Lab to do hands-on practice of the concepts
7. Learn best practices and design patterns for the front-end development
8. Apply the knowledge readily to the next assignment or job

This class assumes no prior experience in the Front-end web design and development concepts. Basic web programming experience is helpful but is not required. Comfort with computers, desktop tools and general computing concepts are expected.

Required Tools and Course Textbooks. No required Textbooks and all materials will be provided.

Recommended Text: (Some reading materials will be provided via pdf files)

- Front-End Web Development: The Big Nerd Ranch Guide, Aquino & Gandee, ISBN- 10: 0134433947 | ISBN-13: 9780134433943
- Web Development and Design Foundations with HTML5, 8/E ISBN-13: 9780134322759
- Fundamentals of Web Development By Randy Connolly, Ricardo Hoar ISBN-10: 0133407152 • ISBN-13: 9780133407150
- AngularJS, JavaScript, and jQuery All in One Brad Dayley, Brendan DayleyISBN-10 0672337428 • ISBN-13: 9780672337420
- Teach Yourself HTML, CSS, and JavaScript All in One, Second Edition, ISBN-13: 978-0-672-33714-7
- Computer Security: Principles and Practice, 3/E, William Stallings and Lawrie Brown ISBN-10: 0133773922 • ISBN-13: 9780133773927
- The Elements of User Experience, By Jesse James Garett ISBN-10: 0321-68368-4
- Basics of Web Design (HTML5 and CSS3), Terry Felke-Morris, ISBN 978-0-13- 397074-6
- Online tutorials on MEAN, Angular6, and Node JS

Course Schedule/Topics Covered:

Class 1	**01/06**	**Web Design using HTML5, CSS3** (HTML Elements, Validation, Web Design Basics, Mobile Web Design, Principles and best practices, CSS Basics, BOX Model and Page Layout Basics) Class Lab
Class 2	**01/13**	**Web Design using HTML5, CSS3** (Element and Class Selectors, CSS page layout manipulation using float, Style a table, Form and controls, HTML5 text form controls, HTML 5 Audio and Video Source Elements, Media Queries, List and Menu creation) Class Lab
Class 3	**01/27**	**Javascript, DataTypes and BOM** (Introduction, Data types, Working with BOM Model , Node) Class Lab
Class 4	**02/03**	**Javascript DOM and Ajax** (Working with DOM Model , Node, Document Object, Element Node Object, Modifying a DOM Element and Properties, xmlHttpRequest object, Debugging Ajax applications) Class Lab
Class 5	**02/10**	**Bootstrap framework for responsive websites** (Mobile First approach, Grid model, Carousal, Wells, Pills, Forms and difference between version 3 and 4) Class Lab
Class 6	**02/17**	**CSS Grid, FlexBox and SASS** Class Lab
Class 7	**02/24**	**Mid Term Exam**

	03/03	**SPRING BREAK – NO CLASSES**
Class 8	**03/10**	**Advanced concepts of Javascript and JQuery** (Understanding DOM Objects, Prototype, and Arrow Functions, Using JQuery Selectors, Navigating and Manipulating DOM Objects using JQuery, Chaining JQuery Object operations, Event Handlers using JQuery and Implementing Callbacks) Class Lab
Class 9	**03/17**	**Introduction to Node JS, Express JS and Mongo DB** (create http server using node, routing using Express JS and Mongo DB concepts) Class Lab
Class 10	**03/24**	**Learning Typescript, React JS** (Syntax, Decision Making, Types, Classes, Objects, Namespaces, Modules, Ambients, Components, Props and Virtual-DOM) Class Lab
Class 11	**03/31**	**React JS** Routing, State, Redux
Class 12	**04/07**	**Angular** (Introduction to Components, Templates, Dependency Injection, Angular setup and first Angular app), Metadata, Directives, Forms, Bindings Class Lab
Class 13	**04/14**	**Final Exam**
Class 14	**04/21**	**Final Project Presentations**

Grading(Relative): Final Grading Weightage

- Assignments 25%
- Quiz 25%
- Midterm Exam 15%
- Final Project and Presentation 20%
- Final Exam 15%

Final Team Project:

A team of 4-5 students will complete the final projects. The project details will be provided during the second/third week of the course. Students are encouraged to choose their topic from healthcare, business, and technology-related fields. Details on project deliverable requirements will be specified at a later time.

Grading Scale

<table>
<tr><td rowspan="2">95-100% A</td><td>87-89.9% B+</td><td>77-79.9% C+</td><td rowspan="3">69.9% or below F</td></tr>
<tr><td>84-86.9% B</td><td>74-76.9% C</td></tr>
<tr><td>90-94.9% A-</td><td>80-83.9%B-</td><td>70-73.9% C-</td></tr>
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Attendance/Late Work Policy

Attendance Policy

Students registered in MGEN courses (INFO, CSYE, and DAMG) are allowed **a maximum of 2 absences per course, with 3 or more absences resulting in an automatic 'F' for that course.** Students are expected to inform their instructors of any absences in advance of the class; if a student is sick long-term or experiences a medical issue that prevents class attendance, it is strongly encouraged that they speak with their Academic Advisor (coe-mgen-gradadvising@northeastern.edu) to learn more about the Medical Leave of Absence. Should a student anticipate being unable to attend 3 or more classes, they should discuss their situation with their Academic Advisor to explore other types of leave in accordance with the University's academic and global entry expectations. International students should review the Office of Global Services webpage to understand their visa compliance requirements.

Teaching Assistants (TAs) or Instructional Assistants (IAs) will be present at each class to collect student attendance.

Late Work Policy

Students must submit assignments by the deadline in the time zone noted in the syllabus. Students must communicate with the faculty prior to the deadline if they anticipate work will be submitted late. Work submitted late without prior communication with faculty will not be graded.

End-of-Course Evaluation Surveys
Your feedback regarding your educational experience in this class is particularly important to the College of Engineering. Your comments will make a difference in the future planning and presentation of our curriculum.

At the end of this course, please take the time to complete the evaluation survey at **https://neu.evaluationkit.com**. Your survey responses are **completely anonymous and confidential**. For courses 6 weeks in length or shorter, surveys will be open one week prior to the end of the courses; for courses greater than 6 weeks in length, surveys will be open for two weeks. An email will be sent to your Northeastern University Mail account notifying you when surveys are available.

Academic Integrity
A commitment to the principles of academic integrity is essential to the mission of Northeastern University. The promotion of independent and original scholarship ensures that students derive the most from their educational experience and their pursuit of knowledge. Academic dishonesty violates the most fundamental values of an intellectual community and undermines the achievements of the entire University.

As members of the academic community, students must become familiar with their rights and responsibilities. In each course, they are responsible for knowing the requirements and restrictions regarding research and writing, examinations of whatever kind, collaborative work, the use of study aids, the appropriateness of assistance, and other issues. Students are responsible for learning the conventions of documentation and acknowledgment of sources in their fields. Northeastern University expects students to complete all examinations, tests, papers, creative projects, and assignments of any kind according to the highest ethical standards, as set forth either explicitly or implicitly in this Code or by the direction of instructors.

Go to http://www.northeastern.edu/osccr/academic-integrity-policy/ to access the full academic integrity policy.

MGEN Student Feedback
Students who would like to provide the MGEN unit with anonymous feedback on this particular course, Teaching Assistants, Instructional Assistants, professors, or to provide general feedback regarding their program, may do so using this survey: https://neu.co1.qualtrics.com/jfe/form/SV_cTlAbH7ZRaaw0Ki

University Health and Counseling Services
As a student enrolled in this course, you are fully responsible for assignments, work, and course materials as outlined in this syllabus and in the classroom. Over the course of the semester if you experience any health issues, please contact UHCS.

For more information, visit https://www.northeastern.edu/uhcs.

Student Accommodations
Northeastern University and the Disability Resource Center (DRC) are committed to providing disability services that enable students who qualify under Section 504 of the Rehabilitation Act and the Americans with Disabilities Act Amendments Act (ADAAA) to participate fully in the activities of the university. To receive accommodations through the DRC, students must provide appropriate documentation that demonstrates a current substantially limiting disability.

For more information, visit https://drc.sites.northeastern.edu.

Library Services
The Northeastern University Library is at the hub of campus intellectual life. Resources include over 900,000 print volumes, 206,500 e-books, and 70,225 electronic journals.

For more information and for education specific resources, visit https://library.northeastern.edu
Network Campus Library Services: Northeastern University Library Global Campus Portals

24/7 Canvas Technical Help
For immediate technical support for Canvas, call 617-373-4357 or email help@northeastern.edu

Canvas Student Resources: https://canvas.northeastern.edu/student-resources/

For assistance with my Northeastern e-mail, and basic technical support:
Visit ITS at https://its.northeastern.edu
Email: help@northeastern.edu
ITS Customer Service Desk: 617-373-4357

Diversity and Inclusion
Northeastern University is committed to equal opportunity, affirmative action, diversity, and social justice while building a climate of inclusion on and beyond campus. In the classroom, members of the University community work to cultivate an inclusive environment that denounces discrimination through innovation, collaboration, and an awareness of global perspectives on social justice.

Please visit http://www.northeastern.edu/oidi/ for complete information on Diversity and Inclusion

Title IX
Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance.

Northeastern's Title IX Policy prohibits Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence, and stalking. The Title IX Policy applies to the entire community, including male, female, transgender students, faculty, and staff.

In case of an emergency, please call 911.

*Please visit **https://www.northeastern.edu/ouec** for a complete list of reporting options and resources both on- and off-campus.*